

GHS Safety Data Sheet

According to Regulation ST/SG/AC.10/30/Rev.8 UN GHS

This Safety Data Sheet cancels and replaces all preceding SDS for this product.

1 IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

RHYTHM
dsm-firmenich product

1.2 Relevant identified uses of the substance or mixture and uses advised against

Fragrance ingredient for use in consumer products.
Not for personal use in this form or concentration.
Intended to be used in the manufacture of products for consumers

1.3 Details of the supplier of the safety data sheet

Firmenich Asia Private Ltd
No 10 Tuas West Road
Singapore 638377
SINGAPORE
Tel: +(65) 6347 2888

P&B.RA.ASIA@dsm-firmenich.com

1.4 Emergency telephone number

+65 3158 1074 NCEC - Available 24/7

2 HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

2.1.1 Classification according to Regulation ST/SG/AC.10/30/Rev.8 [UN/GHS]

Skin Sensitization - Cat. 1	H317
Environmental Hazard (acute) - Cat. 2	H401
Environmental Hazard (chronic) - Cat. 2	H411

2.1.2 Additional information

Full text of listed statements : See section 16

2.2 Label elements

Hazard pictograms:

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Signal Word:

Warning

Hazard Statements:

H317 May cause an allergic skin reaction.
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements:

P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P302+P352 IF ON SKIN: Wash with plenty of soap and water.
P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
P391 Collect spillage.

2.3 Other hazards

No data available at this time.

3 COMPOSITION/INFORMATION ON INGREDIENTS

Mixture of aromatic substances.

Contains :

>= 2.5 < 5.0%
1,3,4,6,7,8-Hexahydro-4,6,6,7,8,8-Hexamethylindeno[5,6-C]Pyran
N° CAS : 0001222-05-5
N° EC: 214-946-9
N° EU REACH: 01-2119488227-29

GHS Classification:

Skin Irritation - Cat. 3 [H316]
Environmental Hazard (acute) - Cat. 1 [H400]
Environmental Hazard (chronic) - Cat. 1 [H410]

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>= 1.0 < 2.5%

Reaction Mass Of: (E)-Oxacyclohexadec-12-En-2-One, (E)-Oxacyclohexadec-13-En-2-One, A) (Z)-Oxacyclohexadec-(12)-En-2-One And B) (Z)-Oxacyclohexadec-(13)-En-2-One

N° CAS : 0111879-80-2

N° EC: 422-320-3

N° EU REACH: 01-0000016883-62

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

>= 1.0 < 2.5%

Cyclopentaneacetic Acid, 3-Oxo-2-Pentyl-, Methyl Ester

N° CAS : 0024851-98-7

N° EC: 246-495-9

N° EU REACH: 01-2119493355-31

Environmental Hazard (acute) - Cat. 2 [H401]

>= 1.0 < 2.5%

Benzyl Salicylate

N° CAS : 0000118-58-1

N° EC: 204-262-9

N° EU REACH: 01-2119969442-31

Skin Sensitization - Cat. 1B [H317]

Eye Irritation - Cat. 2B [H320]

Environmental Hazard (chronic) - Cat. 3 [H412]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

>= 0.5 < 1.0%

Octanal, 2-(Phenylmethylene)-

N° CAS : 0000101-86-0

N° EC: --

N° EU REACH: 01-2119533092-50

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 3 [H316]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 2 [H411]

Acute Toxicity (oral) - Cat. 5 [H303]

>= 0.5 < 1.0%

Hexyl Salicylate

N° CAS : 0006259-76-3

N° EC: 228-408-6

N° EU REACH: 01-2119638275-36

Skin Sensitization - Cat. 1 [H317]

Skin Irritation - Cat. 3 [H316]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

>= 0.5 < 1.0%

Octanal, 7-Hydroxy-3,7-Dimethyl-

N° CAS : 0000107-75-5

N° EC: 203-518-7

Skin Sensitization - Cat. 1B [H317]

Eye Irritation - Cat. 2A [H319]

Environmental Hazard (acute) - Cat. 3 [H402]

>= 0.5 < 1.0%

Naphtho[2,1-B]Furan, Dodecahydro-3a,6,6,9a-Tetramethyl-

N° CAS : 0003738-00-9

N° EC: --

N° EU REACH: 01-2119976315-31

Reproductive Toxicity - Cat. 2 [H361]

Environmental Hazard (chronic) - Cat. 3 [H412]

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>= 0.1 < 0.5%

Hexenyl Salicylate

N° CAS : 0065405-77-8

N° EC: 265-745-8

N° EU REACH: 01-2119987320-37

Reproductive Toxicity - Cat. 2 [H361]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 2 [H411]

Acute Toxicity (oral) - Cat. 5 [H303]

>= 0.1 < 0.5%

Benzenepropanal, .Alpha.-Methyl-4-(1-Methylethyl)-

N° CAS : 0000103-95-7

N° EC: 203-161-7

N° EU REACH: 01-2119970582-32

Reproductive Toxicity - Cat. 2 [H361]

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Environmental Hazard (chronic) - Cat. 3 [H412]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

Flammable Liquid - Cat. 4 [H227]

>= 0.1 < 0.5%

Oxacyclohexadecan-2-One

N° CAS : 0000106-02-5

N° EC: 203-354-6

N° EU REACH: 01-2119987323-31

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 3 [H316]

Environmental Hazard (chronic) - Cat. 2 [H411]

Acute Toxicity (oral) - Cat. 5 [H303]

>= 0.1 < 0.5%

Caryophyllene

N° CAS : 0000087-44-5

N° EC: 201-746-1

N° EU REACH: 01-2120745237-53

Aspiration Hazard - Cat. 1 [H304]

Skin Sensitization - Cat. 1B [H317]

>= 0.1 < 0.5%

Citronellol

N° CAS : 0000106-22-9

N° EC: 203-375-0

N° EU REACH: 01-2119453995-23

Skin Sensitization - Cat. 1B [H317]

Skin Irritation - Cat. 2 [H315]

Eye Irritation - Cat. 2A [H319]

Environmental Hazard (acute) - Cat. 2 [H401]

Acute Toxicity (oral) - Cat. 5 [H303]

Acute Toxicity (dermal) - Cat. 5 [H313]

>= 0.1 < 0.5%

2h-1-Benzopyran-6-ol, 3,4-Dihydro-2,5,7,8-Tetramethyl-2-(4,8,12-Trimethyltridecyl)-

N° CAS : 0010191-41-0

N° EC: 200-412-2

Skin Sensitization - Cat. 1B [H317]

>= 0.1 < 0.5%

1,6,10-Dodecatrien-3-ol, 3,7,11-Trimethyl-

N° CAS : 0007212-44-4

N° EC: 230-597-5

N° EU REACH: 01-2119457636-29

Skin Sensitization - Cat. 1B [H317]

Eye Irritation - Cat. 2B [H320]

Environmental Hazard (acute) - Cat. 1 [H400]

Environmental Hazard (chronic) - Cat. 1 [H410]

4 FIRST-AID MEASURES

4.1 Description of first aid measures

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General information:

As in all cases of potential poisoning, Obtain medical advice immediately.

In case of eye contact:

Irrigate copiously with water for at least 10 minutes. Obtain medical advice if any irritation or evidence of tissue damage persists.

In case of inhalation:

Remove the individual to fresh air and keep at rest. Obtain medical advice immediately.

In case of skin contact:

Remove contaminated clothes. Wash skin with large volumes of water.

If irritation persists, or any sign of tissue damage is apparent, obtain medical advice immediately.

In case of ingestion:

Rinse mouth with water. Obtain medical advice immediately.

4.2 Most important symptoms and effects, both acute and delayed

No information available on the product itself.

4.3 Indication of immediate medical attention and special treatment needed

None known.

5 FIRE-FIGHTING MEASURES

5.1 Extinguishing media

In the event of fire, adequate extinguishers should be used. Avoid inhalation of smoke and fumes. In case of insufficient ventilation, wear suitable respiratory equipment.

5.2 Special hazard arising from the substance or mixture

No specific hazard known.

5.3 Advice for fire-fighters

No specific advice.

6 ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel:

Adequate protective gloves should be worn when handling spillages. No smoking. Avoid naked flames or other potential sources of ignition (eg. electrical equipment).

Avoid skin contamination and inhalation of vapour.

Individual washing routines should be followed after any potential contact.

Ensure adequate ventilation in working areas following accidental releases.

For emergency personnel:

Apply the same recommendations as section 6.1

6.2 Environmental precautions

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Do not discharge directly into drains, air, into soil or into the aquatic environment.

6.3 Methods and material for containment and cleaning up

For containment:

Any absorbent used for cleaning up spillage should be disposed promptly, accordingly to local regulation. Gross spillages should be contained by any means, and disposal of this should be in accordance with Government Regulations.

For cleaning-up:

Spillages should be disposed of in accordance with Governmental Regulations.

7 HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes.

Wear adequate protective gloves protection and eye/face protection.

No smoking. Avoid any sources of ignition.

Avoid exposing to high temperature during processing.

Do not ingest or apply to the skin as such. Good personal washing routines should be followed.

Maintain adequate local and general ventilation where product is handled.

7.1.1 Protective measures

Keep strict control of dust accumulation to a minimum. Maintain adequate local and general ventilation where product is handled. Avoid any sources of ignition.

7.1.2 Advice on general occupational hygiene

Good personal washing routines should be followed.

7.2 Conditions for safe storage, including any incompatibilities

It is good general practice to store in closed, preferably full, containers away from heat sources, and protected from extremes of temperature. Do not re-use the empty container.

Respect general rules for compatibility storage.

7.3 Specific end use(s)

Not available at this time.

8 EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Contains no substances with occupational exposure limit values

Derived No Effect Level (DNEL)

No data available

Predicted No Effect Concentration (PNEC)

No data available

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8.2 Exposure controls

Avoid exposing to high temperature during processing.
Maintain adequate local and general ventilation where product is handled.

8.2.1 Appropriate engineering controls

Maintain adequate local and general ventilation where product is handled and dispensed.

8.2.2 Environmental exposure controls

Not available at this time. Minimize release to the environment.

8.3 Personal protection

Respiratory protection: Where ventilation (LEV) is inadequate to prevent any inhalation of substances, use adequate respiratory apparatus.

Hand protection: Adequate Protective Gloves should be worn.

Eye protection: Adequate safety glasses should be used.

Skin protection: Wear protective clothing, overall if necessary to limit the odour contamination of personal clothing. Individual washing routines should be followed after any potential contact.

9 PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance :	LIQUID
Colour :	COLOURLESS TO PALE YELLOW
Odour :	Characteristic strong odour according to the commercial description of the substance.
Melting point/range (°C) :	Not determined
Initial boiling point/range (°C) :	> 40°C
Flammability :	The product is not flammable.
Explosive properties (St class) :	Not determined
Flash point (closed cup) :	= 102°C
Auto-ignition temperature (°C):	Not determined
Decomposition temperature :	No decomposition if stored and applied as directed.
pH :	Not determined
Water solubility (20°C) :	Not applicable, the product is a mixture
Partition coef. (n-octanol/water) :	Not applicable, the product is a mixture
Vapour pressure (Pa) :	Not determined
Relative density (d 20/20) :	1.004 - 1.014

9.2 Other safety information

None

10 STABILITY AND REACTIVITY

10.1 Reactivity

No reaction known with water.

10.2 Chemical stability

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Good stability at standard temperature.

10.3 Possibility of hazardous reactions

Not known.

10.4 Conditions to avoid

Avoid temperatures above or at least 5 °C below flash point for any flammable liquids.

Do not heat closed containers.

Avoid contact with oxidizing agents.

10.5 Incompatible materials

Not known.

10.6 Hazardous decomposition products

Contact with water or storage under recommended conditions for one year should not produce dangerous decomposition products.

11 TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

This mixture has not been subjected to toxicological testing as an entity. According to available data on the constituents the health classification criteria are met.

(a) acute toxicity

Calculated LD50 Oral: LD50 > 5000 mg/kg bw

(b) skin corrosion/irritation

Not applicable

(c) serious eye damage/irritation

Not applicable

(d) respiratory or skin sensitisation

Skin Sensitization - Cat. 1

(e) germ cell mutagenicity

Not applicable

(f) carcinogenicity

Not applicable

(g) reproductive toxicity

Not applicable

(h) STOT-single exposure



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Not applicable

(i) STOT-repeated exposure
Not applicable

(j) aspiration hazard
Not applicable

12 ECOLOGICAL INFORMATION

This mixture has not been subjected to ecotoxicological testing as an entity. According to available data on the constituents the environmental classification criteria are met.

12.1 Toxicity

Environmental Hazard (acute) - Cat. 2

Environmental Hazard (chronic) - Cat. 2

12.2 Persistence and degradability

No data available

12.3 Bioaccumulative potential

No data available

12.4 Mobility in soil

No data available

12.5 Results of PBT and vPvB assessment

Not applicable

12.6 Other adverse effects

No data available

13 DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Product: The product should be handled according to the instructions given under sections 6, 7 and 8. Dispose of according to local or national regulations. The product should not be allowed to enter drains or the environment

Contaminated packaging: Empty packaging should be disposed according to local or national regulations by an approved waste handling

14 TRANSPORT INFORMATION

In case of accidental spillage or fire during transport, refer to instructions given under points 5, 6, 7 and 8 above.

UNO

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14.1 UN-No: 3082
 14.2 Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
 (Hexahydrohexamethylindenopyran)
 14.3 Class: 9
 14.4 Packing Group: III
 14.5 Environmental hazards: Yes

Land transport (ADR/RID)

14.1 UN-No: 3082
 14.2 Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
 (HEXAHYDROHEXAMETHYLINDENOPYRAN)
 14.3 Class: 9
 14.4 Packing Group: III
 14.5 Environmental hazards: Yes
 14.6 Special precautions for user: Refer to section 7

Sea transport (IMDG-Code)

14.1 UN-No: 3082
 14.2 Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
 (HEXAHYDROHEXAMETHYLINDENOPYRAN)
 14.3 Class: 9
 14.4 Packing Group: III
 14.5 Environmental hazards: Yes
 Marine pollutant: Yes
 14.6 Special precautions for user: Refer to section 7
 14.7 Maritime transport in bulk according to IMO instruments: Not applicable

Air transport (ICAO-IATA)

14.1 UN-No: 3082
 14.2 Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
 (Hexahydrohexamethylindenopyran)
 14.3 Class: 9
 14.4 Packing Group: III
 14.5 Environmental hazards: Not applicable
 14.6 Special precautions for user: Refer to section 7

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14.7 Maritime transport in bulk according to IMO instruments: Not applicable

15 REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:

No relevant information available at this time.

15.2 Chemical Safety Assessment

No data available at this time.

16 OTHER INFORMATION

16.1 Revisions

March-2025 : Version 14.0 - Updates to sections 2, 3, 9, 14, 16

16.2 Key literature references

RIFM database
OECD SIDS
EU IUCLID
Supplier information

16.3 Full text of phrases used under section 2

H317	May cause an allergic skin reaction.
H411	Toxic to aquatic life with long lasting effects.
P273	Avoid release to the environment.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P302+P352	IF ON SKIN: Wash with plenty of soap and water.
P333+P313	If skin irritation or rash occurs: Get medical advice/attention.
P391	Collect spillage.
P363	Wash contaminated clothing before reuse.

16.4 Full text of phrases used under section 3

H227	Combustible liquid
H303	May be harmful if swallowed.
H304	May be fatal if swallowed and enters airways.
H316	Causes mild skin irritation.
H317	May cause an allergic skin reaction.
H320	Causes eye irritation.
H361	Suspected of damaging fertility or the unborn child.
H400	Very toxic to aquatic life.
H401	Toxic to aquatic life.

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H402	Harmful to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H411	Toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.

We believe that the information contained herein is current as of the date of this Safety Data Sheet. Since the use of this information and the conditions of use of the product are not within the control of dsm-firmenich, it is the user's obligation to determine conditions of safe use of the product.

Information contained in this document and related material is proprietary to dsm-firmenich. Disclosure to third parties without the prior written consent of dsm-firmenich except where required by applicable laws and regulations and/or reverse engineering are strictly prohibited.

ORIGINAL CONTROLLED